

Homework (Jalapeno)

- Q1.** The ancient Egyptians used a relation to model the flooding of the Nile River. If the relation is given as $\{(1, 2), (2, 3), (3, 4), (4, 5)\}$, is this relation a function?
- (A) Yes
(B) No
(C) Maybe
(D) Insufficient information
(E) Only for certain values
- Q2.** The ancient Babylonians used a function to calculate the area of a triangle. If the function is given as $f(x) = x \cdot 3$, what is the value of $f(4)$?
- (A) 10
(B) 12
(C) 14
(D) 16
(E) 18
- Q3.** The ancient Greeks used a relation to model the motion of the planets. If the relation is given as a graph, which of the following statements is true about the domain and range?
- (A) Domain is $[-5, 5]$ and range is $[-2, 2]$
(B) Domain is $[-2, 2]$ and range is $[-5, 5]$
(C) Domain is $[-10, 10]$ and range is $[-1, 1]$
(D) Domain is $[-1, 1]$ and range is $[-10, 10]$
(E) Domain is $[-10, 10]$ and range is $[-10, 10]$
- Q4.** The ancient Chinese used a function to calculate the volume of a cylinder. If the function is given as $f(x) = \pi \cdot x^2 \cdot 3$, what is the value of $f(2)$?
- (A) 10π
(B) 12π
(C) 15π
(D) 20π
(E) 25π
- Q5.** The ancient Romans used a relation to model the growth of their empire. If the relation is given as $\{(1, 1), (2, 2), (3, 3), (4, 5)\}$, is this relation a function?
- (A) Yes
(B) No
(C) Maybe
(D) Insufficient information
(E) Only for certain values
- Q6.** The ancient Egyptians used a function to calculate the area of a rectangle. If the function is given as $f(x) = x \cdot 4$, what is the value of $f(3)$?
- (A) 10
(B) 12
(C) 14
(D) 16
(E) 18
- Q7.** The ancient Babylonians used a relation to model the motion of the sun. If the relation is given as a table, which of the following statements is true about the domain and range?
- (A) Domain is $[-5, 5]$ and range is $[-2, 2]$
(B) Domain is $[-2, 2]$ and range is $[-5, 5]$
(C) Domain is $[-10, 10]$ and range is $[-1, 1]$
(D) Domain is $[-1, 1]$ and range is $[-10, 10]$
(E) Domain is $[-10, 10]$ and range is $[-10, 10]$
- Q8.** The ancient Greeks used a function to calculate the volume of a sphere. If the function is given as $f(x) = 4 \frac{\pi \cdot x^3}{3}$, what is the value of $f(1)$?
- (A) $\frac{2}{3} \cdot \pi$
(B) $\frac{4}{3} \cdot \pi$
(C) $\frac{5}{3} \cdot \pi$
(D) $\frac{6}{3} \cdot \pi$
(E) $\frac{8}{3} \cdot \pi$

Open Ended Questions (FRQ)

- Q9.** The ancient Chinese used a function to model the population growth of their cities. If the function is given as $f(x) = 2x + 5$, find the domain and range of this function and evaluate $f(2)$.
- Q10.** The ancient Romans used a relation to model the trade routes of their empire. If the relation is given as $\{(1, 2), (2, 3), (3, 4), (4, 5)\}$, determine if this relation is a function and find its domain and range.

Chef's Tips

- Check for the vertical line test
- Ensure proper function notation
- Verify the domain and range of each function